

Oil price and US dollar exchange rate: Change detection of bi-directional causal impact

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ABSTRACT

The purpose of the paper is to detect the changes in the causal relationship between international oil prices and the US dollar real effective exchange rate, using the Shi et al. (2018, 2020) approach. The proposed recursive evolving methods allow the identification of the causal change in the oil – exchange rate nexus – and of precise episodes of causality and instability in this relationship. Our main findings show that the oil prices Granger-cause the US dollar exchange rate starting with 2005, whereas the intensity of the causality increases in the aftermath of the 2008–2009 Global Financial Crisis. In addition, the strength of the causality fluctuates during crisis times. At the same time, we discover that the US dollar exchange rate Granger-causes the oil prices during the 2008–2009 Global Financial Crisis. These findings are robust to the way the real effective exchange rate is computed, and are robust as to the use of an alternative oil price index or to the influence of economic policy uncertainty. The results provide important information to policymakers and international investors, showing the difficulty of predicting with accuracy the shock transmission between oil prices and the US dollar during high turbulence episodes. In addition, our findings reveal that oil and commodity markets offer smaller and smaller risk-hedging opportunities.

1. Introduction

The relationship between exchange rates and international oil prices is intensively investigated in the literature, whereas a considerable portion of papers analyzes the connection between the US dollar (USD) exchange rate and crude oil prices (for a recent literature review, please refer to Beckmann et al., 2020).¹ The two-way causality between these variables is explained in earlier papers by different transmission channels and the potential observed co-movement of variables. Nevertheless, immediately after the outburst of the 2008–2009 Global Financial Crisis (GFC), the co-movement becomes less clear given the high fluctuations recorded by international oil prices (Beckmann et al., 2020). Moreover, as Kilian and Zhou (2019) underline, it is necessary to distinguish between co-movements and the underlying causality between exchange rates and oil prices. Therefore, recent empirical investigations employ different categories of time-series nonlinear approaches to understand

the connection between these variables.

As Beckmann et al. (2020) show, one group of studies resorts to Granger-type analyses and focuses on short-term response of oil prices (exchange rates) to shocks in exchange rates (oil prices) resorting to Vector Autoregression (VAR) frameworks (e.g., Iwayemi and Fowowe, 2011; Ji et al., 2020). A part of these studies makes the distinction between the short- and long-run interconnection between oil prices and exchange rates and employs various cointegration analyses (e.g., Chaudhuri and Daniel, 1998). A second group of studies uses different nonlinear approaches designed to investigate the co-movement and dependence structure between variables, such as time-frequency approaches (e.g., Tiwari et al., 2013), copula functions (e.g., Reboredo, 2012; Ji et al., 2019a) or volatility spillover approaches such as Generalized Autoregressive Conditional Heteroskedasticity (GARCH) processes (e.g., Ding and Vo, 2012) and dynamic spillover frameworks (e.g., Albuлесcu et al., 2019). Most of these studies highlight the

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¹ It is widely recognized that oil prices represent one of the main sources of shocks to currency markets (Bal and Rath, 2015; Basher et al., 2012; Ji et al., 2015, 2020). At the same time, given that international oil prices are denominated in USD, the exchange rate fluctuations pass-through the oil prices. This makes extremely important the investigation of the connection between oil prices and USD exchange rate for policymakers, central bankers and international investors.

intensification of the oil price-exchange rates nexus in specific periods.

However, it is well-known that Granger-type analyses are sensitive to the estimation period, while the results are influenced by homoskedasticity/heteroskedasticity conditions. At the same time, the second category of approaches deals with co-movement investigations rather than causality analyses, and does not offer enough predictability as classic Granger causality tests do. In this context, we add to the existing work by applying a new time-varying causality approach following Shi et al. (2018, 2020), to investigate the causal relationship between USD real exchange rate and crude oil prices. Compared to other time-varying causality approaches, this methodology is different in several ways. First, this approach does not necessitate differencing or detrending the data as it adopts robust econometric methods for different integration and cointegration properties of variables. Second, this new testing method permits to date economic turbulences and to detect possible real-time switch in causal relationship directions between variables. Third, the test is able to locate precisely the origination and collapse dates of every causality episode. Fourth, the methodology is similar to the recent approaches for financial bubble detection, and the test is easily implemented both in homoskedasticity and conditional heteroskedasticity conditions. Therefore, it has a good positive detection rate and allows for early and effective policy interventions. Fifth, the technique can be adopted for different methods that address the time-varying nature of the causal relationship, such as the forward expanding-window test (e.g., Swanson, 1998), the Markov-switching Granger causality approach (Psaradakis et al., 2005), or the rolling-window Granger causality test (e.g., Balcilar et al., 2010). Shi et al. (2018, 2020) show that the recursive evolving method they propose offers a better finite sample performance compared with other existing methods. We also use the rolling-window test for comparison purpose.

We also contribute to the existing studies by performing different robustness analyses for the time-varying causal relationship between USD real exchange rate (USD REER) and crude oil prices (West Texas Intermediate – WTI), covering the period 12 April 1996 to 17 April 2020.² Firstly, we use two indicators for the USD REER, namely the REER computed in a broad form (60 economic partners), and the REER calculated in a narrow form (27 economic partners). Secondly, we consider the Brent Crude Index (BRENT) instead of WTI index as a proxy for international crude oil prices. Thirdly, we remove the influence of US economic policy uncertainty (US EPU) on international oil prices. Indeed, a set of previous studies documents the interactions between oil prices and the US EPU index computed by Baker et al. (2016). In this line, Aloui et al. (2016), Antonakakis et al. (2014), Kang and Ratti (2013), Kang and Wang (2018), Roubaud and Aroui (2018) and Shahzad et al. (2019) show that oil price shocks impact the EPU in both oil-exporting and oil-importing countries. At the same time, Albuлесcu et al. (2019) underline the role of US EPU in influencing the oil price – exchange rates nexus whereas Akram (2020) assesses the impact of geopolitical uncertainty. Therefore, in a separate robustness check we orthogonalize the oil prices on US EPU and we afterwards reassess the time-varying causality between oil prices and the USD REER.

Our paper is close to Wen et al. (2018), who examine the time-varying causality among crude oil prices and USD exchange rate, using alternative tests. Like Wen et al. (2018) we discover a one-way causal relationship running from crude oil prices to USD exchange rate. However, different from Wen et al. (2018), we notice that the causality intensifies after 2012. Moreover, we also discover a short

period of causality running from the USD REER to WTI prices during the 2008–2009 GFC.

In what follows, we present the literature review (Section 2), the data and Shi et al. (2018) methodology (Section 3). The main empirical results are presented in Section 4, whereas Section 5 addresses the robustness checks. The last section of the paper concludes.

2. Literature Review

2.1. Theoretical Transmission Channels Between Oil Prices and Exchange Rates

Several theories describe the connection between oil price and exchange rate. Most of them underline how oil prices pass through exchange rates. In this line, Darby (1982) describes the supply-demand channel, which relates oil price fluctuations to higher inflation. As a response to inflationary pressures, the domestic interest rate adjusts accordingly, thus affecting the value of the domestic currency, and in agreement with the real interest rate parity hypothesis. In addition, the portfolio and wealth channel (the balance of payment theory) described by Golub (1983) and by Krugman (1983) shows that oil price fluctuations determine a wealth transfer across oil-importing and oil-exporting countries. More precisely, as Bénassy-Quéré et al. (2007) state, both the short-run portfolio channel and the medium- and long-run wealth channels determine a currency appreciation for the oil-exporting countries, generating a currency depreciation of oil-importing ones. Further, the terms of the trade channel (international trade theory) introduced by Amano and Van Norden (1998a, 1998b) states that the connection between oil prices and real exchange rates is explained by the terms of trade and by the purchasing power parity, both driven by arbitrage forces. Whereas the balance of payment theory focuses on the nominal exchange rates, the international trade theory investigates the oil price effects on the real exchange rate.

The exchange rate impact on international oil prices is equally plausible given that oil prices are usually denominated in USD. The low of one price shows that the USD appreciation dampens the demand for oil outside the US (Bloomberg and Harris, 1995). In addition, the USD appreciation might generate an increase of oil supply, triggering a decrease of oil prices at the international level (Coudert et al., 2008). In fact, Fratzscher et al. (2014) show that oil producers either increase the oil prices or cut supply if the USD depreciates. Finally, Zhang et al. (2008) and several years later Beckmann and Czudaj (2013) identify a risk spillover from the USD exchange rate to international oil prices, in the context of commodity-markets financialization.

Therefore, the theoretical elements point in favor of a bidirectional causality between oil prices and exchange rates, whereas the transmission mechanisms are easily identified for countries that adopt free-floating exchange rate arrangements. Nevertheless, even in the case of intermediary exchange rate regimes (Lv et al., 2018), or in the case of implementation of exchange rate targeting regimes (Reboredo, 2012), the linkages between oil prices and exchange rates cannot be ignored.

Beckmann et al. (2020) show that the general level of prices is the key variable that connects oil prices and exchange rates. Therefore, the nexus oil prices – exchange rates might be influenced by a series of exogenous shocks, associated with the level of inflation, as the monetary policy, the output growth, the economic uncertainty, or the stock prices. In this line, the monetary key rate responds to oil price shocks to avoid an increase of inflation, thus impacting the exchange rate. At the same time, periods of economic boom generate a higher demand for oil, triggering an increase of oil prices. However, the effect on the exchange rate is ambiguous in this case. On the one hand, in the context of an increased interest rate as a response to a positive output gap (see, for example, the Taylor rule), the uncovered interest rate parity indicates a depreciation of the domestic currency. On the other hand, the overshooting models point in favor of a short-run currency appreciation. The uncertainty represents another key variable that influences the oil prices

² We use the REER instead of nominal exchange rate to assess the relationship between international oil prices and USD exchange rate, in agreement with the terms of trade channel introduced by Amano and Van Norden (1998a, 1998b). Clearly, the REER allow us to consider the price dynamics and the role of inflation (Bénassy-Quéré et al., 2007). Previous papers resorting to the REER are those of Cashin et al. (2004), Chen and Chen (2007), De Vita and Trachanas (2016), Kilian (2009) or Tiwari et al. (2013).

– exchange rates nexus. With a focus on the US, [Albuлесcu et al. \(2019\)](#) underline that uncertainty related to the monetary policy influences the global financial cycle. Therefore, uncertainty facilitates the volatility of interactions between commodity and financial markets ([Miranda-Agrippino and Rey, 2015](#)). In this context, investors' risk preferences change and allow for shocks transmission between oil and exchange rate markets, which are highly liquid. Several empirical studies consider such exogenous shocks when the link between oil and exchange rate markets are assessed (e.g., [Albuлесcu et al., 2019](#); [Degiannakis et al., 2018](#); [Kilian and Zhou, 2019](#)).

2.2. Empirical Studies

Early empirical papers that investigate the connection between oil prices and exchange rates use either linear VAR frameworks (e.g., [Huang and Guo, 2007](#); [Ding and Vo, 2012](#); [Iwayemi and Fowowe, 2011](#); [Turhan et al., 2013](#)), structural VAR ([Fueki et al., 2020](#); [Ji et al., 2020](#)), or cointegration approaches (e.g., [Chaudhuri and Daniel, 1998](#); [Rautava, 2004](#); [Zhang et al., 2008](#); [Lizardo and Mollick, 2010](#)), and make the distinction between short- and long-run effects.

More recent papers resort to non-linear approaches to assess the connection between oil and currency markets ([Akram, 2004](#)). Likewise, [Mensah et al. \(2017\)](#) investigate the 2008–2009 GFC' impact on the oil price – exchange rate nexus for a set of oil-dependent economies. The authors discover that the long-run relationship is more stable in the aftermaths of the crisis. With a focus on emerging countries, [Lin and Su \(2020\)](#) perform a similar investigation. The authors decompose the oil price and exchange rate series and resort to an autoregressive distributed lag (ARDL) model. Their unique findings show that the shocks in oil price have a different impact on currency markets in oil-importing compared with oil-exporting countries, and the influence is significant only in the long run. In the same spirit, [Jung et al. \(2020\)](#) use a nonlinear ARDL model and test the Granger causality among oil price and the Canada-US exchange rate. They report a bi-directional long-run causality whereas in the short run, they report an asymmetric relationship, in line with [Jiang et al., 2020](#). Other studies resort to regime-switching models ([Basher et al., 2016](#); [Roubaud and Arouri, 2018](#); [Youssef and Mokni, 2020](#)), bivariate mixture models ([Xu et al., 2019](#)), conditional causality approaches ([Bouoiyour et al., 2015](#)), or detrended cross-correlation analyses ([Reboredo et al., 2014](#); [Jiang and Gu, 2016](#)), and report mixed findings.

The literature also shows that the oil price – exchange rate nexus is characterized by different frequency cycles. Therefore, several papers use wavelets to investigate this relationship, an approach that allows the localization of co-movements both in time and frequency ([Tiwari et al., 2013](#); [Uddin et al., 2013](#)). For example, [Tiwari and Albuлесcu \(2016\)](#) use a continuous wavelet approach and a battery of asymmetric Granger-causality tests and assess the connection between the Indian Rupee and the USD exchange rate. Different from other papers, they report intense co-movements in the medium-run (2–4-year cycle). In addition, they report that the Rupee-USD exchange rate Granger-causes oil prices in the long run, whereas an opposite result is noticed in the short run. Performing a comparative investigation between oil-importing and oil-exporting countries within a time-frequency framework, [Yang and Cai \(2017\)](#) discover a homogenous oil- currency markets connection only in the case of oil-exporting countries, and around the 2008–2009 GFC.

Starting with [Reboredo \(2012\)](#), several papers assess the dependence structure characterizing the oil prices – exchange rate nexus. With a focus on the bilateral exchange rate between the US and China, [Ji et al. \(2019a\)](#) apply a battery of time-varying copula models and report that the extreme dependence among oil prices and the China-US exchange rate is positive and manifests at lower tails, while the volatility spillover running from oil to exchange rate is asymmetric. [Bedoui et al. \(2019\)](#) use a nested copula model to assess the dependence structure between oil, gold, and the USD exchange rate. In line with most previous papers, the authors discover that the dependence between commodity and currency

markets increase in crisis times. At the same time, [Chkir et al. \(2020\)](#) observe that oil prices may only represent a weak hedge against exchange rates. Opposite findings are reported by [Dai et al. \(2020\)](#), who combine copula with wavelets and use a wavelet-based vine-copula approach to assess the dependence among oil prices, gold prices, and the USD exchange rate. They report a stronger dependence in the long run compared with the short-term horizon.

The asymmetries that characterize the oil-exchange rate relationship are usually investigated using quantile regressions. For example, [Jiang et al. \(2020\)](#) resort to a quantile-on-quantile approach and a causality analysis to investigate the influence of oil-price shocks on exchange rates for a set of developed and developing countries. Whereas for developed countries the causality in quantiles manifests at all quantiles, in the case of developing countries this relationship is not significant at middle quantiles. Similar findings are reported by [Nusair and Olson \(2019\)](#). In addition, the authors discover that the asymmetric effects are different in intensity between bullish and bear currency markets.

A distinctive group of papers performs spillover analyses between oil prices and exchange rates. Between the first papers in this area is the work by [Zhang et al. \(2008\)](#) who search for the spillover effect of USD exchange rate over the international oil prices and show that a depreciation of the US dollar is a key driver of international oil prices. In a different setting, [Albuлесcu et al. \(2019\)](#) investigate the spillover effect of oil prices on commodity currencies and assess to what extent the US economic policy uncertainty influences this relationship. With a focus on China and India, [Ahmad et al. \(2020\)](#) document positive spillover from oil to currency markets from 2013 to 2019. The dynamic connectedness analysis performed by [Malik and Umar \(2019\)](#) shows an intensification of the oil – currency market relationship after 2009, whereas oil-price shocks do not explain exchange rate volatility in the case of oil-dependent countries. A directional analysis considering the oil prices and emerging markets exchange rates is conducted by [Baghestani et al. \(2019\)](#) for the period 1994–2017. The findings show that oil prices predict the REER direction of change, except for China. A different approach is adopted by [Nouira et al. \(2019\)](#) who resort to an asymmetric non-causality specification to investigate the volatility spillover among oil prices and exchange rates. Considering the series' returns, the authors report significant spillovers running from oil to currency markets from Middle East and North Africa (MENA) countries.

Given the manifestation of oil-price shocks and currency markets dynamics, most recent research states that oil price – exchange rate nexus presents features that are time varying (e.g., [Wen et al., 2018, 2019](#); [Huang et al., 2020](#)). [Wen et al. \(2018\)](#) resort to a time-varying parameter structural VAR (TVP-VAR) model and report a nonlinear Granger causality running from the oil price to USD, whereas the opposite does not apply. Using the same empirical approach, [Wen et al. \(2019\)](#) examine the combined effect of different financial factors, including the USD exchange rate, on the international oil prices. First, the authors report a stronger impact after the 2008–2009 GFC. Second, their findings show that the USD exchange rate is the dominant factor of oil prices before the crisis. Running a time-varying analysis, [Huang et al. \(2020\)](#) discover that unexpected oil price shocks have a stronger impact on the main exchange rates compared to general oil price shocks. We add to this bulk of literature and we apply [Shi et al.'s \(2018, 2020\)](#) approach to investigate the connection among oil prices and the USD REER. Different from the TVP-VAR framework, [Shi et al.'s \(2018, 2020\)](#) method relies on a recursive evolving framework that allows to isolate changes in causal relationships based on a Wald test.

3. Data and Methodology

3.1. Data

We use the daily data of the oil price and the USD REER calculated in a broad form (60 economic partners), and in a narrow form (27 economic partners) for robustness purposes.³ The data range from 12 April 1996 to 17 April 2020, and are extracted from the Bank for International Settlements (BIS) and the US Energy Information Administration (EIA) for the US dollar exchange rates and the oil prices, respectively. Series are investigated under log return form $r_t = \log(p_t/p_{t-1})$ where p_t denotes the time series price at day t .

The variables under investigation are presented in Fig. 1. The figure shows that variables exhibit a linear and upward (downward) trend with an explosive behavior, which recommends the use of a time-varying framework.

3.2. Methodology

The causality relationship between the USD exchange rate and crude oil prices is investigated using the recent time-varying causality procedure proposed by Shi et al. (2018, 2020), supposing the use of recursive evolving algorithms. The rolling-window Wald test causality approach, which represents a particular case of the recursive evolving approach, is used for comparison purposes.⁴ To present this procedure, let us introduce the lag augmented VAR advanced by Dolado and Lütkepohl (1996). According to Dolado and Lütkepohl (1996), a Granger causality test for a potential integrated variable y_t may be conducted if the VAR model is written as:

$$Y = \tau\Gamma' + X\Theta' + B\Phi' + \varepsilon \quad (1)$$

where $Y = (y_1, \dots, y_T)_{T \times n'}$, $\tau = (\tau_1, \dots, \tau_T)_{T \times 2'}$, $\tau_t = (1, t)_{2 \times 1'}$, $X = (x_1, \dots, x_T)_{T \times np'}$, $x_t = (y_{t-1}', \dots, y_{t-p}')$, $\Theta = (\beta_1, \dots, \beta_p)_{n \times np}$, $B = (b_1, \dots, b_T)_{T \times nd'}$, $b_t = (y_{t-p-1}', \dots, y_{t-p-d}')$, $\Phi = (\beta_{p+1}, \dots, \beta_{p+d})_{n \times nd}$, $\varepsilon = (\varepsilon_1, \dots, \varepsilon_T)_{T \times n'}$, and d represents y_t maximum order of integration.

The null hypothesis $H_0 : R\theta = 0$ imposes some restrictions yielding the Walt test as follow:

$$w = [R\hat{\theta}]' [R(\hat{\Omega} \otimes (X'QX)^{-1})R']^{-1} [R\hat{\theta}] \quad (2)$$

where $\hat{\theta} = \text{vec}(\hat{\Theta})$ is the row vectorization, $\hat{\Theta}$ represents a least-squares estimator of type $\hat{\Theta} = X'QX(X'QX)^{-1}$, $\hat{\Omega} = T^{-1}\hat{\varepsilon}\hat{\varepsilon}'$ and R is a $m \times n^2p$ matrix where m denotes the restrictions' number. Under the assumption of conditional homoscedasticity and the null hypothesis, the Wald statistic is asymptotically χ_m^2 .

The Shi et al. (2018, 2020) novel real-time-varying causality approach supposed the use of supremum (sup) Wald statistic sequences. Therefore, we note with $W_{f_2}(f_1)$ the Wald statistic over $[f_1, f_2]$, where the sample size fraction is denoted by $f_w = f_2 - f_1 \geq f_0$. The sup Wald statistic is given by:

$$SW_f(f_0) = \frac{\sup_{(f_1, f_2) \in \Lambda_0, f_2 = f} \{W_{f_2}(f_1)\}}{(f_1, f_2) \in \Lambda_0, f_2 = f} \quad (3)$$

where, for reduced sample size $f_0 \in (0, 1)$, $\Lambda_0 = \{(f_1, f_2) : 0 < f_0 + f_1 \leq f_2 \leq 1, \text{ and } 0 \leq f_1 \leq 1 - f_0\}$.

Thus, as Shi et al. (2018) state, the rolling-window procedure becomes a special case of the recursive evolving procedure. Based on simulation investigations, Shi et al.'s (2018) reported that the power of the rolling and recursive procedures increases when f_0 rises from 0.18 to

0.24. In line with Shi et al.'s (2018), we adopt a window $f_0 = 0.2$ as an optimal choice that ensures enough observations to issue the regression.

The dating rules for a simple switch case are represented by the following procedures:

- recursive evolving

$$\hat{f}_e = \frac{\inf_{f \in [f_0, 1]} \{f : SW_f(f_0) > scv\}}{f \in [\hat{f}_e, 1]} \{f : SW_f(f_0) < scv\} \quad (4)$$

- rolling

$$\hat{f}_e = \frac{\inf_{f \in [f_0, 1]} \{f : W_f(f - f_0) > cv\}}{f \in [\hat{f}_e, 1]} \{f : W_f(f - f_0) < cv\} \quad (5)$$

where $\hat{f}_e$ and $\hat{f}_t$ represent, respectively, the first observation (chronological estimated), for which the test statistics exceed or fall below the critical values in the causal relationship. Critical values refer to the origination and termination points, respectively. The critical values of the statistics W_f and SW_f are cv and scv , respectively. Also, the origination and termination dates for multiple switches are computed in a similar way.

4. Empirical Findings

To implement the time-varying causality (TVC) tests, we use the Bayesian information criteria (BIC) with a maximum of 12 lags to estimate the VAR model. Following Shi et al. (2018), we use the minimum size f_0 equal to 0.2, which includes 1205 observations. Likewise, we use a bootstrapping procedure with 499 replications to identify the critical values, whereas the empirical size, controlled over a three-year period, is established to 5%.

Before conducting the TVC tests, we check the stationarity of the log-return variables and discover that all variables are stationary in level, i. e., $I(0)$.⁵ As a result, the time-varying causality using the recursive evolving and rolling-window algorithms is conducted by setting to zero the additional lag parameter d (for additional explanations, please refer to Shi et al., 2018).

For the causal connection between WTI and USD REER, the time-varying Wald test statistics, and their bootstrapped critical values, are shown in Fig. 2. Each column in Fig. 2 illustrates the direction of causality running from USD REER to oil prices and vice-versa. The two rows in each figure refer to test statistics' sequences, acquired from the recursive-evolving and rolling-window methods, respectively.

The TVC effects that run from the USD REER (broad) to WTI are shown in the first graph of Fig. 2. We observe a short episode of causality for both the recursive evolving (Fig. 2a) and the rolling-window (Fig. 2c) procedures. The TVC between USD REER and WTI is localized around the 2008–2009 Global Financial Crisis. During this period, the oil prices recorded huge jumps and drops. Apparently, after a long period of USD depreciation recorded in 2000s, the USD appreciation in the second part of 2008 contributed to the drop in international oil prices in 2009. Indeed, according to the low of one price, the USD depreciation triggered an increase in international oil prices before the crisis. After the outburst of the 2008–2009 GFC, the USD unregistered a real appreciation and the global demand contracted, a situation which caused a drop

³ Also, for robustness purposes we use Brent crude instead of WTI prices.

⁴ The empirical performance algorithms presented by Shi et al.'s (2018, 2020) following their simulation experiments indicate better performances for the recursive evolving window procedure compared with other approaches.

⁵ Unit root results are available upon request.

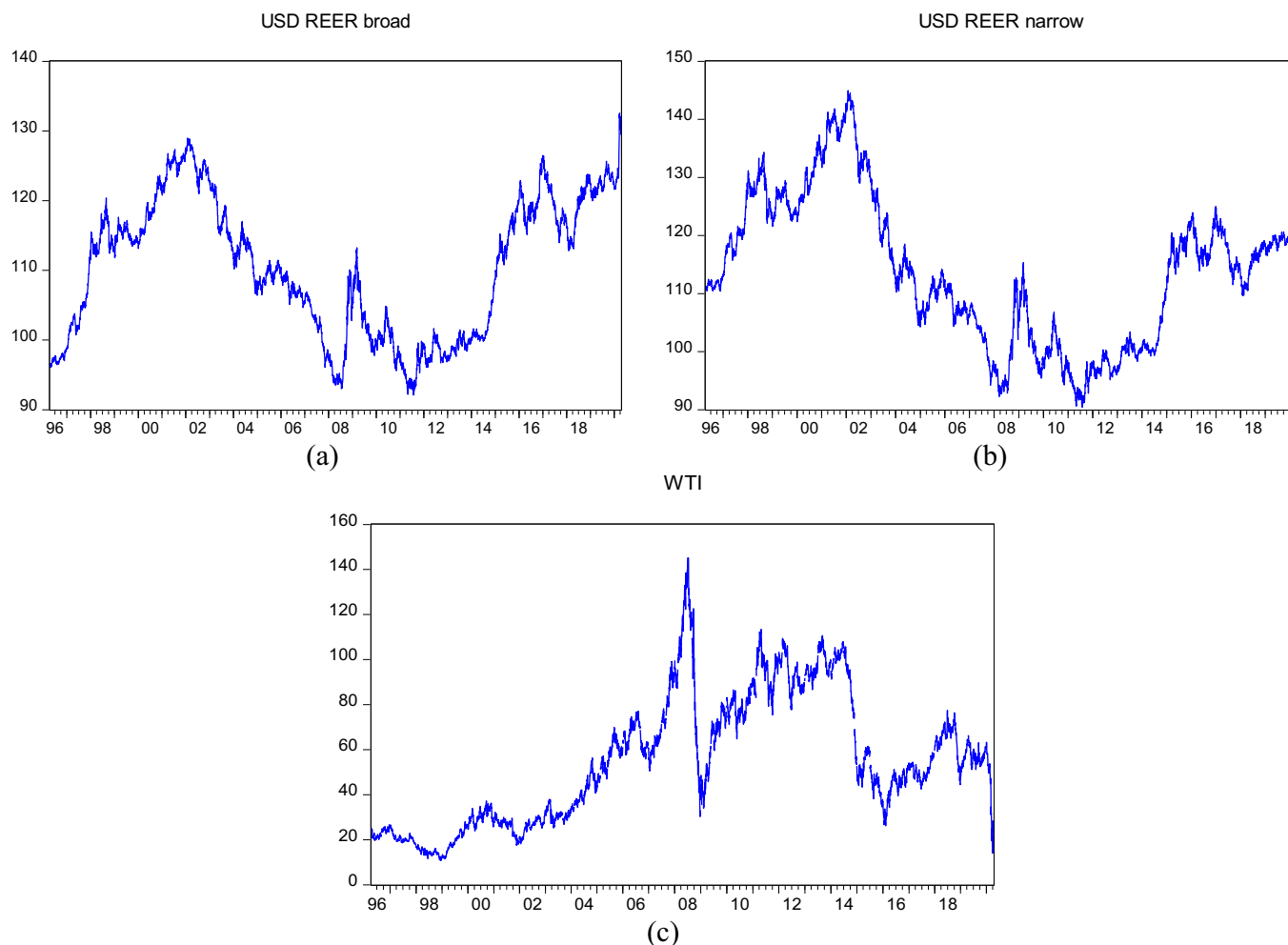


Fig. 1. Time series plot of the variables.

in international oil prices. In addition, given the theory of commodity-markets financialization (Beckmann and Czudaj, 2013), risk spillovers between commodity and financial markets increase in crisis times (Ji et al., 2019b).

On the one hand, our results show the intensification of oil price – exchange rate interactions, in line with the findings reported by Bedoui et al. (2019). On the other hand, our result contradicts the findings reported by Wen et al. (2018), obtained in a similar TVC setting, which show a one-way causality from oil to the USD exchange rate. Our findings are in line with those reported by Jung et al. (2020), who discover a bi-directional causality between oil price and USD exchange rate.

However, in line with most previous papers (e.g. Jiang et al., 2020; Wen et al., 2018), we clearly identify a TVC running from WTI to USD REER for the entire period between 02 May 2006 and 17 April 2020. This result is very significant, and it is confirmed by both the recursive evolving (Fig. 2b) and the rolling-window (Fig. 2d) procedures. Both approaches indicate an intensification of the causality following the 2008–2009 GFC, findings in agreement with those reported by Malik and Umar (2019), Wen et al. (2019) and Ji et al. (2020). If we consider the recursive evolving results (Fig. 2(b)) that have better statistical performances according to Shi et al. (2018), we also notice a stabilization of the causality at a high level for the last three years. An original result shows that the causality running from WTI to USD REER fluctuates during the time span from 2008 to 2009. In fact, in 2008, the causality increases, showing that the oil prices led to an appreciation of the USD. However, the drop in oil prices caused, in a less strong way, the depreciation of USD installed for a short period in 2009 (see Fig. 1a, b).

5. Robustness Analyses

5.1. Alternative Specification for the USD REER

Our findings might be influenced by the way the USD REER is computed. Indeed, Beckmann et al. (2020) show that causality results between oil prices and exchange rates are influenced by sample choices and the way the exchange rate is computed. Therefore, in the first step we check the robustness of the main results, using the narrow approach for the USD REER calculation (Fig. 3).

Fig. 3(a,c) indicate the same short episode of causality from the USD exchange rate to the oil prices during the 2008–2009 GFC. Another short episode of significant causality is noticed in 2020, in the context of the new coronavirus pandemic, with a small breach between 23 and 25 March 2020. For the other way of causality, we report a significant TVC starting with March 2006. The rolling-window approach indicates small breaks in the first part of 2006 (Fig. 3(c)). Similarly, the recursive evolving procedure finds that the WTI Granger-causes the USD REER over the period from 15 March 2006 to 17 April 2020, a result which confirms the main findings reported in Fig. 2.

5.2. Alternative Specification for the International Oil Prices

In a second robustness-check analysis, we use the log return of the Brent crude price index to investigate the TVC with USD REER. The results for the USD REER (broad measure) are displayed in Fig. 4, whereas the findings for the USD REER (narrow measure) are presented

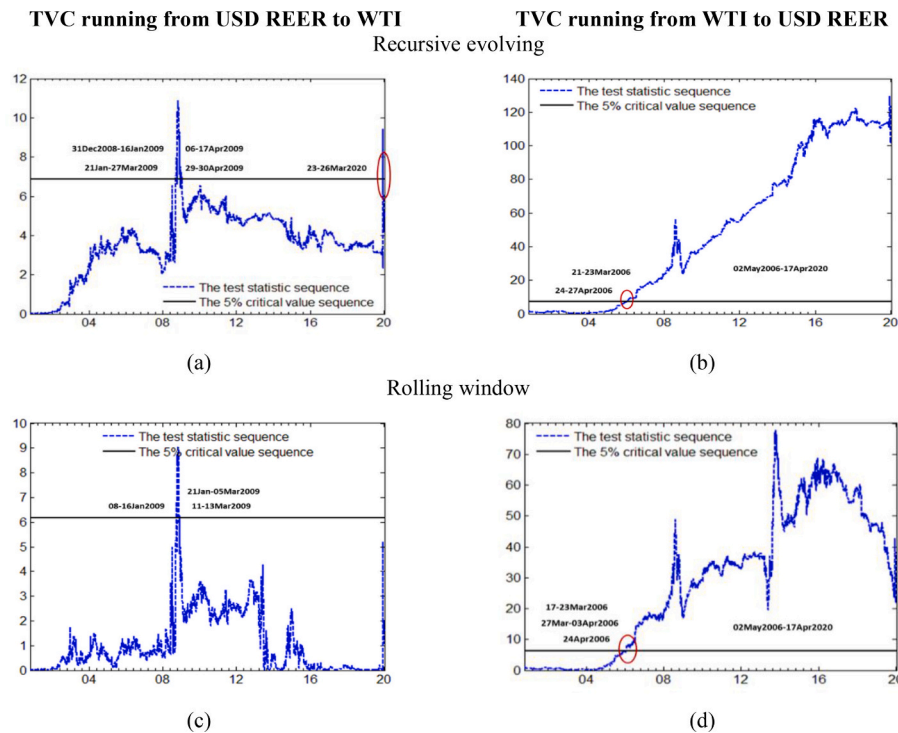


Fig. 2. TVC tests: WTI vs. USD REER (broad).

Notes: (i) on the vertical axes we have the Wald test statistics whereas the sequence of the test statistics appears on the horizontal axis, (ii) the red circle shows the change detection in the causal relationship, (iii) if the blue dotted line is situated above the 5% significance level (black line), the causal relationship is significant.

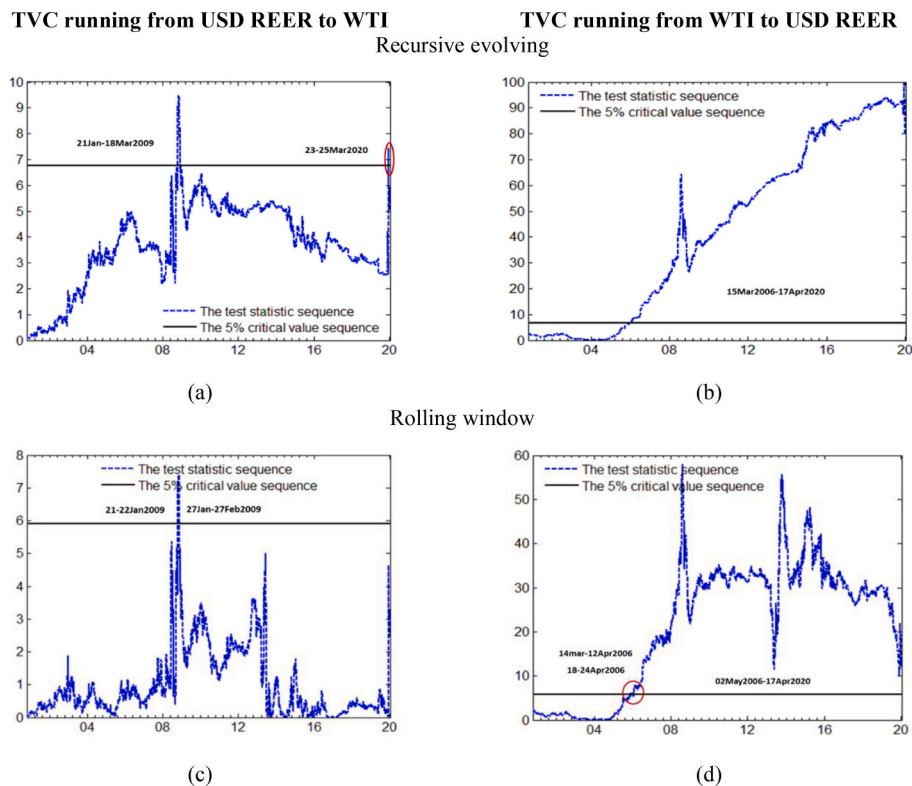


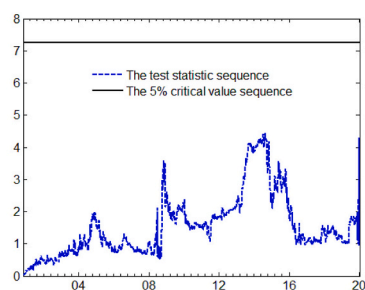
Fig. 3. TVC tests: WTI vs. USD REER (narrow).

Notes: (i) on the vertical axes we have the Wald test statistics whereas the sequence of the test statistics appears on the horizontal axis, (ii) the red circle shows the change detection in the causal relationship, (iii) if the blue dotted line is situated above the 5% significance level (black line), the causal relationship is significant.

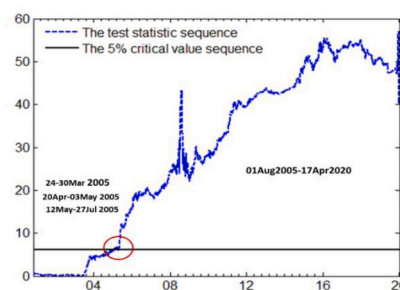
TVC running from USD REER to BRENT

TVC running from BRENT to USD REER

Recursive evolving

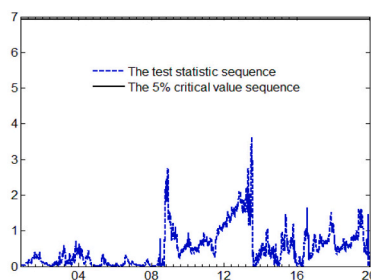


(a)

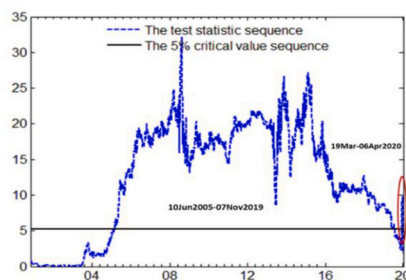


(b)

Rolling window



(c)



(d)

Fig. 4. TVC tests: BRENT vs. USD REER (broad).

Notes: (i) on the vertical axes we have the Wald test statistics whereas the sequence of the test statistics appears on the horizontal axis, (ii) the red circle shows the change detection in the causal relationship, (iii) if the blue dotted line is situated above the 5% significance level (black line), the causal relationship is significant.

in Fig. A1 (Appendix).

Looking to the first graph of Fig. 4, we notice that the test statistics are below the critical values. This observation is valid for the entire sample period, meaning that the null of no causality running from USD REER to Brent crude cannot be rejected. Indeed, the Brent crude and the WTI oil prices were closely connected until the outburst of the 2008–2009 GFC, whereas the difference increased afterwards, with the Brent crude being priced higher. This element might explain the lack of significant causal effect that runs from the USD exchange rate to oil prices during the 2008–2009 period.

Using the recursive evolving procedure to assess the TVC running from Brent crude to USD REER (Fig. 4b), we find evidence of significant causality episodes from 24 March to 27 July 2005 with small breaks, and from 01 August 2005 to 17 April 2020. As a result, we cannot reject the null hypothesis of no Granger causality. This result confirms the main findings of this paper and are in line with those reported by Zhang (2013), showing that oil price and exchange rate dynamics are subject to structural breaks. At the same time, if we consider the results of the rolling-window procedure, we notice that the test statistics concerning the predictive power of the Brent crude on the USD REER are over their critical values during the period between 10 June 2005 to 07 November 2009, and over 19 March - 09 April 2020. It appears that during the start of the COVID-19 crisis when the oil prices recorded their lowest level in March and April 2020, the causality was no longer significant. These findings remain unchanged if the narrow approach is used for the USD REER computation (Fig. A1).

5.3. Controlling for the Influence of US EPU

The uncertainty induced by governmental policies influences the financial markets volatility (Pástor and Veronesi, 2012). At the same time, it is well-known that the US monetary policy drives the global financial cycle (Passari and Rey, 2015). In this context, the US EPU

might impact the oil price – exchange rate relationship (Albulescu et al., 2019). Therefore, in a separate robustness check we control for the effect of US EPU on international oil price. We orthogonalize the WTI oil prices on the US EPU (Baker et al., 2016) and we obtain a new oil price series (WTI*), free of EPU influence. These findings presented in Fig. 5 confirms the main results, either we use the recursive evolving or the rolling window approach to assess the time-varying causality. Indeed, the causality running from the USD REER to the oil prices is only significant during the 2008–2009 GFC. At the same time, the causality from oil prices to the USD REER becomes significant after 2005 and is increasing, with some fluctuations during the crisis.

6. Conclusions

The relationship between international oil prices and the USD exchange rate is characterized by nonlinearities and asymmetries, documented by recent empirical papers. Nevertheless, these papers use complex approaches to detect co-movements and dependencies between oil and currency markets, which do not allow the identification of the causal relationship between variables, thus affecting the predictability capacity of investors and authorities. Moreover, the several recent papers that use TVC methods either identify break points in this relationship, or resort to rolling-window approaches, which affect the capacity of identification of the change in causal relationships between oil prices and USD exchange rate.

To overcome this limitation, we implement Shi et al.'s (2018, 2020), approach that relies on a recursive evolving procedure which allows the identification of the causal change in the oil – exchange rate nexus, and identifies the instability periods in the TVC relationship. The precise dates of any episode of causality are identified, which makes it easy for policymakers to understand the shock transmission between oil and currency markets.

Our main findings show that the oil prices Granger-cause the USD

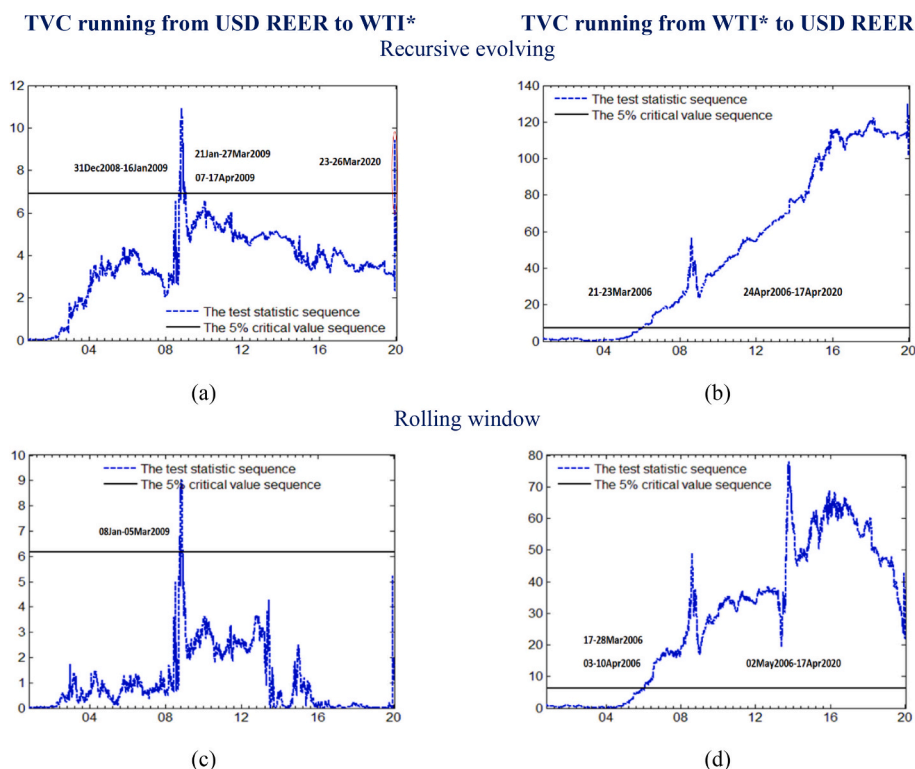


Fig. 5. TVC tests: WTI* vs. USD REER (broad).

Notes: (i) on the vertical axes we have the Wald test statistics whereas the sequence of the test statistics appears on the horizontal axis, (ii) the red circle shows the change detection in the causal relationship, (iii) if the blue dotted line is situated above the 5% significance level (black line), the causal relationship is significant, (iv) WTI* is the WTI orthogonalized on US EPU index.

exchange rate starting with 2005, whereas the intensity of the causality increases in the aftermath of the 2008–2009 Global Financial Crisis. In addition, according to the recursive evolving approach, the strength of the causality fluctuates during the period 2008 to 2009, while the oil price increases recorded in 2008 have a powerful effect on the USD appreciation. The rolling-window approach presents similar findings, except for the last period, when the strength of the causality diminishes. At the same time, we notice that the USD REER Granger-causes the oil prices during the 2008–2009 GFC. These findings are robust to the way the USD REER is computed and also to the use of an alternative oil-price index (i.e., Brent crude). However, when the Brent crude is used instead of WTI, the causality running from the USD REER to oil prices during the crisis is no longer significant. This result can be explained by the fact that the Brent crude is the representative price index in Europe. At the same time, although the previous literature documents the important role played by the US EPU on the oil price – exchange rate nexus, we show that the level of economic policy induced uncertainty does not bring essential changes in the time-varying documented causal relationship. As a matter of fact, the causality intensity fluctuates during the 2008–2009 GFC, when the US EPU is very high. However, we clearly show that the causal effect between oil prices and the USD REER intensifies after the crisis, pointing in the favor of more integrated commodity and financial markets. At the same time our results confirm the

increasing role of oil in influencing the global economic system.

Although our results bring noteworthy clarifications to the causal relationship between oil and USD exchange rate and the approach, we use is easy to follow, the extreme dependences between oil prices and USD REER are not considered, which represents a limitation of this approach. Certainly, during high turbulence episodes, the investigation of shock spillovers and the appearance of extreme event deserve more attention (see for example Albuлесcu et al., 2020; Bouri et al., 2018).

The policy implications of our results are the following. First, during the crisis episodes, both investors and policymakers should be aware of the fact that the causal relationship between oil prices and USD fluctuates, which makes it harder to estimate the price shocks pass-through from one market to another. Second, it is worthy to note that oil and commodity markets offer smaller risk-hedging opportunities during the last period. Thirdly, the central bankers should be aware about the increasing role of international oil prices in influencing currency markets.

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Appendix A. Appendix

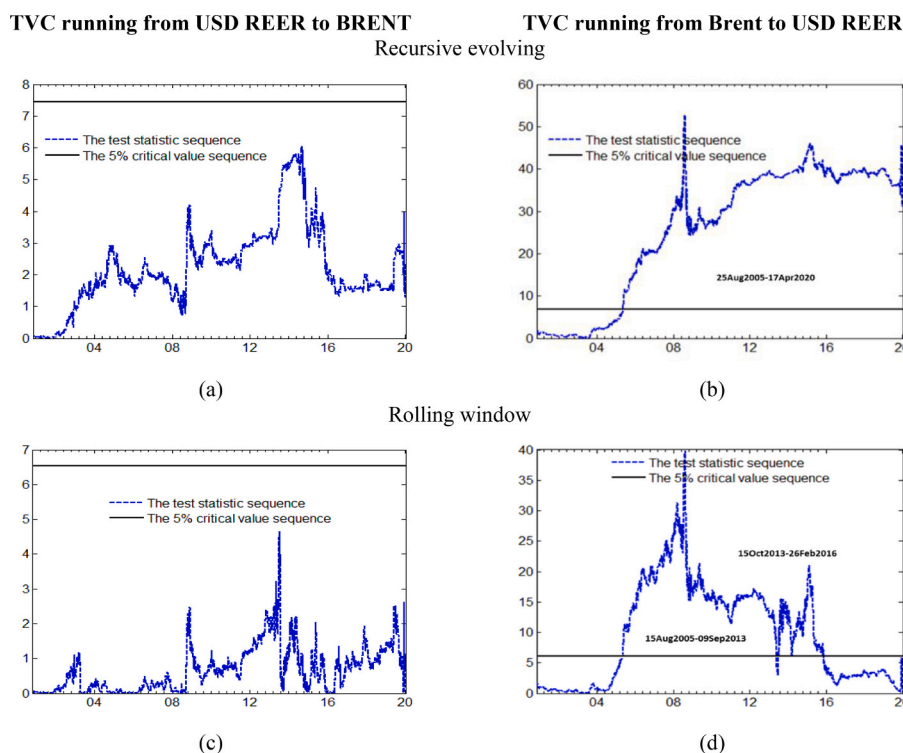


Fig. A1. TVC tests: Brent vs. USD REER (narrow).

Notes: (i) on the vertical axes we have the Wald test statistics whereas the sequence of the test statistics appears on the horizontal axis, (ii) the red circle shows the change detection in the causal relationship, (iii) if the blue dotted line is situated above the 5% significance level (black line), the causal relationship is significant.

Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eneco.2021.105385>.

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